



Database Management Systems

Relational Operations

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Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor

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A procedural language that consists of a set of operations, where each operation takes one or two relations as input and produces a new relation as the result.

- The result of an operation is a new relation, which is formed from one or more input relations.
- This property makes the algebra "closed" (all objects in relational algebra are relations)
- These relations can be further manipulated using operations of the same algebra.
- A **sequence of relational algebra** operations forms a **relational algebra expression**.
- The result of a relational algebra expression is also a relation that represents the result of a database query.

Operators:

Unary Operators:

Select: σ

Project: Π

Rename: ρ

Binary Operators:

Union: $\cup$

Intersection: $\cap$

Set difference: $-$

Cartesian product: $\times$

Join: $\bowtie$

Relational
Algebra
Operators



Use this table for query examples:

University Database : **Consider Instructor relation** shown below:

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Database Management Systems

Example-2 Company Database State



EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

- The **SELECT** operation is used to choose a subset of the tuples from a relation that satisfies a **selection condition**.
- We can consider the SELECT operation to be a filter that keeps only those tuples that satisfy a qualifying condition.

Notation: $\sigma_p(r)$

- p is called the **selection predicate**
- The SELECT operation can also be visualized as a horizontal partition of the relation into two sets of tuples—those tuples that satisfy the condition and are selected, and those tuples that do not satisfy the condition and are filtered out.



- **Example 1:** Select those tuples of the *instructor* relation where the instructor is in the “Physics” department.

Query: $\sigma(\text{Dept_Name} = \text{“Physics”})(\text{Instructor})$

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
33456	Gold	Physics	87000

- **Example 2:** Select the tuples for all employees who either work in department 4 and make over \$25,000 per year, or work in department 5 and make over \$30,000.

Query: $\sigma(\text{Dno}=4 \text{ AND } \text{Salary}>25000) \text{ OR } (\text{Dno}=5 \text{ AND } \text{Salary}>30000)(\text{EMPLOYEE})$

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

SELECT Operation Properties

- The SELECT operation σ ($\langle \text{condition} \rangle$) produces a relation S that has the same schema (same attributes) as R
- SELECT is commutative: $\sigma_{\langle \text{condition1} \rangle}(\sigma_{\langle \text{condition2} \rangle}(R)) = \sigma_{\langle \text{condition2} \rangle}(\sigma_{\langle \text{condition1} \rangle}(R))$
- Because of commutative property, A cascade (sequence) of SELECT operations may be applied in any order:

or

$$\sigma_{\langle \text{cond1} \rangle}(\sigma_{\langle \text{cond2} \rangle}(\sigma_{\langle \text{cond3} \rangle}(R))) = \sigma_{\langle \text{cond2} \rangle}(\sigma_{\langle \text{cond3} \rangle}(\sigma_{\langle \text{cond1} \rangle}(R)))$$

$$\sigma_{\langle \text{cond1} \rangle}(\sigma_{\langle \text{cond2} \rangle}(\sigma_{\langle \text{cond3} \rangle}(R))) = \sigma_{\langle \text{cond1} \rangle \text{ AND } \langle \text{cond2} \rangle \text{ AND } \langle \text{cond3} \rangle}(R))$$

- We allow comparisons to be used in the selection predicate.
 $=, \neq, >, \geq, <, \leq$
- We can combine several predicates into a larger predicate by using the connectives:
 $\wedge$ (**and**), $\vee$ (**or**), $\neg$ (**not**)

Example: Find the instructors in Physics with a salary greater \$90,000.

$$\sigma_{dept_name="Physics" \wedge salary > 90,000} (instructor)$$

Example: Fetch all the details of the 'Research' department.

$$\sigma_{dept_name="Research"} (department)$$

NOTE: The number of tuples in the result of a *SELECT* is less than (or equal to) the number of tuples in the input relation *R*.

PROJECT(Π)

- A unary operation that returns its argument relation, with certain attributes left out.

Notation: $\Pi_{A_1, A_2, A_3 \dots A_k}(r)$

where $A_1, A_2, \dots, A_k$ are attribute names and r is a relation name.

- The PROJECT operation, selects certain columns from the table and discards the other columns.
- **Duplicate rows removed from result, since relations are sets**
- The PROJECT operation can be visualized as a vertical partition of the relation
- The list of specified columns (attributes) is retrieved and other attributes in each are discarded.

- **Example 1:**

Eliminate the dept_name attribute of instructor or select only id, name and salary of the

instructor

Query:

$\Pi_{ID, name, salary}(instructor)$

Result:

ID	name	salary
10101	Srinivasan	65000
12121	Wu	90000
15151	Mozart	40000
22222	Einstein	95000
32343	El Said	60000
33456	Gold	87000
45565	Katz	75000
58583	Califieri	62000
76543	Singh	80000
76766	Crick	72000
83821	Brandt	92000
98345	Kim	80000



- Example 2: Fetch the Lname, Fname, Salary information about the Employees.

Query:

$\Pi_{Lname, Fname, Salary}(EMPLOYEE)$

- Example 3: Fetch the Sex and Salary information about the Employees.

Query:

$\Pi_{Sex, Salary}(EMPLOYEE)$

Sex	Salary
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

output for example 3

Lname	Fname	Salary
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

output for example 2

Properties of PROJECT operator:

- The project operation removes any duplicate tuples.
- This is because the result of the project operation must be a set of tuples.
- Mathematical sets do not allow duplicate elements.
- The number of tuples in the result of projection $\Pi(R)$ is always less or equal to the number of tuples in R.
- If the list of attributes includes a key of R, then the number of tuples in the result of PROJECT is equal to the number of tuples in R.

PROJECT is *not* commutative

- $\pi_{\langle \text{list1} \rangle} (\pi_{\langle \text{list2} \rangle} (R)) = \pi_{\langle \text{list1} \rangle} (R)$ as long as $\langle \text{list2} \rangle$ contains the attributes in $\langle \text{list1} \rangle$

- The result of a relational-algebra operation is a relation and therefore of relational-algebra operations can be composed together into a **relational-algebra expression**.

Example: Find the names of all instructors in the Physics department.

$$\Pi_{name}(\sigma_{dept_name = "Physics"}(instructor))$$

Instead of giving the name of a relation as the argument of the PROJECT operation, we give an expression that evaluates to a relation.

- We may want to apply several relational algebra operations one after the other.
- Either we can write the operations as a single relational algebra expression by nesting the operations, or we can apply one operation at a time and create intermediate result relations.

- To retrieve the first name, last name and salary of all employees who work in the department 5, we must apply a select and a project operation.
- We can write a single relational algebra expression as follows:

$$\Pi_{Fname, Lname, Salary}(\sigma_{DNo = 5}(Employee))$$

- Or we can explicitly show the sequence of operations, giving a name to each immediate relation:

$$DEP5_EMPS \leftarrow \sigma_{DNO=5}(Employee)$$

$$RESULT \leftarrow \Pi_{Fname, Lname, Salary}(DEP5_EMPS)$$

- The results of relational-algebra expressions do not have a name that we can use to refer to them. The rename operator, ρ , is provided for that purpose.
- The RENAME operator is denoted by ρ (rho)

- The expression:

$$\rho_x(E)$$

returns the result of expression E under the name x

- Another form of the rename operation:

$$\rho_{x(A1,A2, \dots, An)}(E)$$

- In some cases, we may want to rename the attributes of a relation or the relation's name or both.
- Useful when a query requires multiple operations

- The general RENAME operation r can be expressed by any of the following forms:

$r_{S(B_1, B_2, \dots, B_n)}(R)$ changes both:

- the relation name to S , *and*
- the column (attribute) names to $B_1, B_1, \dots, B_n$

$r_S(R)$ changes:

- the *relation name* only to S

$r_{(B_1, B_2, \dots, B_n)}(R)$ changes:

- the *column (attribute) names* only to $B_1, B_1, \dots, B_n$

Alternate notation to rename only few attributes:

$$\rho_{OldName \rightarrow NewName}(r)$$

- For Example, $\rho_{Father \rightarrow Parent}(Paternity)$. For the given operation, the attribute “Father” will be renamed as “Parent” for the relation Paternity.
- If there are two or more attributes involved in renaming operation, then ordering is more meaningful.

$$\rho_{Brach, Salary \rightarrow Location, Pay}(Employee)$$

- RESULT will have the *same attribute names* as DEP5_EMPS (ie same attributes as EMPLOYEE)

$\text{RESULT} \leftarrow \pi_{\text{FNAME, LNAME, SALARY}}(\text{DEP5_EMPS})$

- The 10 attributes of DEP5_EMPS are *renamed* to F, M, L, S, B, A, SX, SAL, SU, DNO, respectively

Note: the symbol(<- is an assignment operator discussed in next slide)

$\text{RESULT (F, M, L, S, B, A, SX, SAL, SU, DNO)} \leftarrow$
 $\rho \text{ RESULT (F.M.L.S.B,A,SX,SAL,SU, DNO)}(\text{DEP5_EMPS})$

- It is convenient at times to write a relational-algebra expression by assigning parts of it to temporary relation variables.
- The assignment operation is denoted by $\leftarrow$ and works like assignment in a programming language.
- Example: **Fetch the FirstName, LastName and Salary of all employees who work in department 5.**

```
DEP5_EMPS  $\leftarrow$   $\sigma_{DNO=5}$ (EMPLOYEE)  
RESULT  $\leftarrow$   $\pi_{FNAME, LNAME, SALARY}$  (DEP5_EMPS)
```

- With the assignment operation, a query can be written as a sequential steps consisting of a series of assignments followed by an expression whose value is displayed as the result of the query.

Example:

(a)

Fname	Lname	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

$TEMP \leftarrow \sigma_{Dno=5}(EMPLOYEE)$

$R(First_name, Last_name, Salary) \leftarrow \pi_{Fname, Lname, Salary}(TEMP)$

(b)

TEMP

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston,TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston,TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

Figure 8.2

Results of a sequence of operations. (a) $\pi_{Fname, Lname, Salary}(\sigma_{Dno=5}(EMPLOYEE))$.

(b) Using intermediate relations and renaming of attributes.

- **Scenario:** You have a table Employees with columns EmployeeID, Name, Department, Salary, and JoiningYear (eg:2016).

Question: How would you create a query to find the names and departments of employees who joined before 2020, but without showing the EmployeeID and Salary columns?

- **Scenario:** A database contains a Customers table with fields CustomerID, FirstName, LastName, City, and PurchaseAmount.

Question: If you need to rename the City column to Location, write the relational query for this scenario.

- **Scenario:** You are working with a table Products with columns ProductID, ProductName, Category, Price, and StockQuantity.

Question: How would you select the ProductName and Price of all products that are in the Electronics category and have more than 10 items in stock?

- **Scenario:** You have a table Employees with columns EmployeeID, Name, Department, Salary, and JoiningYear (eg:2016).

Question: Write a query to find the names and departments of employees who joined before 2020, but without displaying the EmployeeID and Salary columns?

Answer: $\Pi_{\text{Name, Department, JoiningYear}}(\sigma_{\text{JoiningYear} < 2020}(\text{Employees}))$

- **Scenario:** A database contains a Customers table with fields CustomerID, FirstName, LastName, City, and PurchaseAmount.

Question: You need to rename the City column to Location, write the relational query for this scenario.

Answer: $\rho_{\text{City} \rightarrow \text{Location}}(\text{Customers})$

- **Scenario:** You are working with a table Products with columns ProductID, ProductName, Category, Price, and StockQuantity.

Question: How would you select the ProductName and Price of all products that are in the Electronics category and have more than 10 items in stock?

Answer: $\Pi_{\text{ProductName, Price}} (\sigma_{\text{Category='Electronics' } \wedge \text{ StockQuantity} > 10} (\text{Products}))$



THANK YOU

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